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Original Paper

Freely Available Arabic Corpora: A Scoping Review

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ABSTRACT

Background: Corpora play a vital role when training machine learning (ML) models and building systems that use natural language processing (NLP). It can be challenging for researchers to access corpora in a language other than English, and even more so if the corpora are not available for free of cost. The Arabic language is used by more than 1.5 billion Muslims and is the native language of over 250 million people as the Quran, the core text of Islam, is written in Arabic.
Objective: To highlight peer-reviewed literature reporting free and accessible Arabic corpora. We aimed to benefit researchers by providing insights into freely available Arabic and accessible corpora, allowing them to achieve their research goals with ease.
Methods: By conducting a scoping review using PRISMA guidelines, we searched the most common information technology (IT) databases and identified free of cost and accessible Arabic corpora.
Results: We identified a total of 48 accessible corpora sources available free of cost in the Arabic language, we present our findings according to categories to further help readers understand the corpora with direct links where available. The results were classified by corpora type into five categories based on their primary purpose.
Conclusion: Arabic is underrepresented considering freely available corpora as most such corpora are available in English. Although previous studies have performed searches for corpora, ours is the first of its kind as it follows the PRISMA guidelines and includes peer-reviewed articles in the literature, obtained by searching the most common IT databases and source recommendations from language experts.

1. Introduction

Corpora are a large collection of documents normally stored as digital texts, which often contain tagged metadata comprising additional information. For any artificial intelligence (AI)-built system, regardless of the complexity of the algorithm, having an acceptable sized dataset is crucial. A previous study performed searches using online queries to locate freely available Arabic corpora and repositories [1]. It reported challenges regarding the extraction of information, such as inaccessibility and disorganization on personal and research group websites. In the same study [1] the authors report how the Arabic resources section in The European Network of Excellence in Human Language Technologies contained only 23 entries with most of them being over 12 years old. Furthermore, they report other global efforts no longer being maintained despite huge efforts to build such repositories. Globally, the Arabic language, used by more than 1.5 billion Muslims and is the native language of over 250 million people as the Quran, the core text of Islam, is written in Arabic [2]. However, freely available corpora in the Arabic language remain relatively scarce, which can hinder the progress of AI-driven research, especially considering natural language processing (NLP) techniques and their increase usage in many fields. In a previous study, [1] the authors attempted to locate freely available Arabic corpora and repositories by performing an online search and surveying 20 researchers. However, this study did not consider previous studies for identifying other sources. It is valuable to assess previous studies as authors often report the source of the Arabic corpora.

In addition to the lack of studies exploring accessible corpora in the Arabic language of those that do exist [1], none followed PRISMA guidelines that include a search of the published literature. Moreover, high costs of corpora of up to $6000 per license can be a major hindrance to accessibility [2]. Identification of corpora size is also important especially for AI and NLP projects, a larger corpus is beneficial owing to

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its diversity and representation of the language being studied; therefore, researchers suggest that one million words are sufficient for general linguistic research [3]. Thus, it is crucial to collate available Arabic corpora that have been reported in the literature (as opposed to relying on web searches), are free of cost, accessible and contain a diverse and largely sized corpus.

To the best of our knowledge, no previous reviews have searched the literature to identify accessible Arabic corpora within the criteria. Therefore, the aim of this study was to identify free of cost and accessible Arabic corpora using a scoping review. There are no up-to-date reviews, besides the Zaghouani study [1] and this alone was as a strong motivation for conducting this review

2. Methods

We explored the freely available Arabic corpora via a systematic method to establish the current size and nature of the corpora, and to identify its scope. We followed the Preferred Reporting Items for Systematic Review and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) as a guideline to perform a comprehensive review [4].

2.1. Search strategy

2.1.1. Search sources

Several primary bibliographic databases were searched, including ScienceDirect, Springer, IEEE Xplore, an ACM Digital Library, and LREC database. The most common unpaid database, arXiv was also searched. Furthermore, we retrieved the first 100 citations by conducting a search on Google Scholar, which generally yields numerous studies based on the searched keywords. These searches were conducted between the 3rd and 15th February 2021. The reference list of the included studies (according to our inclusion criteria) was examined so that further relevant studies using backward reference list checking could be retrieved. Forward reference list checking was also conducted.

2.1.2. Search terms

We used a previous study on freely available Arabic corpora [1] as a guide for the format of this scoping review. We highlighted 2 groups of search terms to be inclusive and relevant, the first group represented the Arabic language, which included: “Arabic” OR “Arab” OR “Arabian”. The second group represented the corpora, which included terms such as, “Corpora” OR “Corpus” OR “Dataset” OR “Lexical database*” OR “Lexical Dataset*” OR “Multi Dialect*” OR “Lexicon*”. Both groups were combined as per the example from the IEEE database query: (“Arabic” OR “Arab” OR “Arabian”) AND (“Corpora” OR “Corpus” OR “Lexical database” OR “Lexical Dataset” OR “Multi Dialect” OR “Lexicon”).

2.2. Study eligibility criteria

The main inclusion criteria were that the Arabic corpora should be free and accesible in text or speech format. We excluded: a) any paid text or speech in the Arabic corpora/corpus, b) free Arabic corpora/corpus that are inaccessible, incomplete, or contains missing information such as size or working website; c) peer-reviewed articles, dissertations, conference proceedings, workshop papers, and reports. We excluded reviews, conference abstracts, and editorials. Because the followed baseline survey was published in 2014 and covered several free Arabic corpora [1], we decided to search for studies published between 2015 and 2021 to perform a more up-to-date review. No restrictions were applied regarding the country of publication, the source of the corpora, or type of the used annotation method.

2.3. Study selection

In the first phase of study screening duplicate studies were removed. In the second phase, two reviewers (NA and MS) individually screened

Table 1
Table showing study numbers coded for referencing purposes.

Study no.	Title and reference
S2	1.5 billion words Arabic corpus [2]
S4	A 700M+ Arabic corpus: KACST Arabic corpus design and construction [3]
S6	Habibi-a multi dialect multinational Arabic song lyrics corpus [6]
S7	ASAYAR: A Dataset for Arabic-Latin Scene Text Localization in Highway Traffic Panels [7]
S8	PAAD: Political Arabic articles dataset for automatic text categorization [8]
S9	An Arabic Multi-source News Corpus: Experimenting on Single-document Extractive Summarization [9]
S10	A fine-grained Arabic dialect identification dataset [10]
S11	A new hybrid method for Arabic multi-font text segmentation, and a reference corpus construction [11]
S12	Hate and offensive speech detection on Arabic social media [12]
S13	Stance prediction and claim verification: An Arabic perspective [13]
S14	Sentiment Analysis for Arabic Reviews using Machine Learning Classification Algorithms [14]
S15	A7′ ta: Data on a monolingual Arabic parallel corpus for grammar checking [15]
S16	OSIAN: Open source international Arabic news corpus-preparation and integration into the CLARIN-infrastructure [16]
S17	Studying the history of the Arabic language: language technology and a large-scale historical corpus [17]
S18	Building a Speech Corpus based on Arabic Podcasts for Language and Dialect Identification [18]
S19	Dzdc12: a new multipurpose parallel Algerian Arabizi–French code-switched corpus [19]
S20	Arsentd-lev: A multi-topic corpus for target-based sentiment analysis in Arabic Levantine tweets [20]
S21	KUNUZ: A Multi-Purpose Reusable Test Collection for Classical Arabic Document Engineering [21]
S22	Arap-tweet: A large multi-dialect twitter corpus for gender, age and language variety identification [22]
S23	Arabic word analogies and semantics of simple phrases [23]
S24	Sentiment lexicon for sentiment analysis of Saudi dialect tweets [24]
S25	MoArLex: An Arabic sentiment lexicon built through automatic lexicon expansion [25]
S26	Arabic senti-lexicon: Constructing publicly available language resources for Arabic sentiment analysis [26]
S27	Building the Classical Arabic Named Entity Recognition Corpus (CANERCorpus) [27]
S28	An open corpus for the computational research of Arab-Andalusian music [28]
S29	A monolingual parallel corpus of Arabic. Procedia computer science [29]
S30	The madar Arabic dialect corpus and lexicon [30]
S31	Curras: an annotated corpus for the Palestinian Arabic dialect [31]
S32	Tashkeela: Novel corpus of Arabic vocalized texts, data for auto-diacritization systems [32]
S33	Towards intelligent Arabic text-to-speech application for disabled people [33]
S34	ANT corpus: an Arabic news text collection for textual classification [34]
S35	Arasenti-tweet: A corpus for Arabic sentiment analysis of Saudi tweets [35]
S36	Shamela: A large-scale historical Arabic corpus [36]
S37	An Arabic-Hebrew parallel corpus of TED talks [37]
S38	SmartATID: A mobile captured Arabic Text Images Dataset for multi-purpose recognition tasks [38]
S39	Nileulex: A phrase and word level sentiment lexicon for Egyptian and modern standard Arabic [39]
S40	A large scale corpus of Gulf Arabic [40]
S41	Word embeddings for Arabic sentiment analysis [41]
S42	Machine translation experiments on PADIC: A parallel Arabic dialect corpus [42]
S43	Improving coverage of rule based NER systems [43]
S44	Arsentd-lev: A multi-topic corpus for target-based sentiment analysis in Arabic Levantine tweets [20]
S45	Stance prediction and claim verification: An Arabic perspective [13]
S46	Cairo Student Code-Switch (CSCS) Corpus: An Annotated Egyptian Arabic-English Corpus [44]
S47	A spelling correction corpus for multiple Arabic dialects [45]
S48	Daict: A dialectal Arabic irony corpus extracted from twitter [46]
S49	Dart: A large dataset of dialectal Arabic tweets [47]
S50	A corpus of Levantine Arabic dialects [48]

(*continued on next page*)

Table 1 (*continued*)

Study no.	Title and reference
S51	The WAW Corpus: The First Corpus of Interpreted Speeches and their Translations for English and Arabic [49]
S52	A morphologically annotated corpus of Emirati Arabic [50]
S53	Sentiment lexicons for Arabic social media [51]
S54	Open Islamicate Texts Initiative (OpenITI) [52]
S55	Shamela: A large-scale historical Arabic corpus [36]
S56	Wit3: Web inventory of transcribed and translated talks [53]
S57	A parallel corpus for statistical machine translation [54]
S58	The united nations parallel corpus v1. 0 [55]
S59	Error Detection for Arabic Text Using Neural Sequence labeling [56]
S60	Toward a Broad-coverage Bilingual Corpus for Speech Translation of Travel Conversations in the Real World [57]
S61	Linguistic Data Consortium University of Pennsylvania [58]
S62	ALTID: Arabic/Latin text images database for recognition research [59]

Table 2
General characteristics of the studies.

Characteristic	Studies	Study reference (refer to Table 1)
Corpus name	Yes: 38	[S4],[S6]–[S11],[S13]–[S16],[S18]–[S22], [S24]–[S28],[S30]–[S36],[S38]–[S40], [S42],[S44]–
	No: 10	[S46]–[S51] [S2],[S12], [S17], [S23], [S29], [S37], [S41], [S43], [S52], [S53]
Publication type	Conference paper: 31	[S6], [S10], [S14], [S16], [S18], [S21]–[S25], [S27]–[S30], [S33]–[S35], [S38]–[S43] [S46]–[S53]
	Journal article: 12	[S4], [S7], [S8], [S9],[S11], [S12], [S15], [S17], [S19], [S26], [S31], [S32]
	Workshop: 3	[S13], [S36], [S37]
	Preprint: 2	[S2], [S44]
Year	2020: 12	[S6]– [S14],[S46]-[S48]
	2019: 7	[S15]–[S21]
	2018:12	[S22]–[S30], [S49],[S50],[S52]
	2017: 5	[S31]–[S35]
	2016: 8	[S2], [S36]–[S41],[S53]
	2015: 3	[S4], [S42], [S43]
Country	Qatar: 5	[S22], [S30],[S48],[S49],[S51]
	USA: 6	[S10], [S17], [S36], [S41], [S44], [S45]
	Italy: 1	[S37]f
	UAE: 3	[S40],[S47],[S52]
	KSA: 6	[S2], [S4], [S15], [S24], [S33], [S35]
	Algeria: 4	[S18], [S19], [S32], [S42]
	Palestine: 1	[S31]
	Spain: 1	[S28]
	Tunisia: 5	[S9], [S21], [S34], [S38], [S43],
	Malaysia: 2	[S26], [S27]
	Morocco: 3	[S7], [S11], [S16]
	Egypt: 3	[S14], [S25], [S39]
	Czech: 1	[S23]
	Canada: 2	[S12],[S53]
	Oman: 1	[S29]
	UK: 1	[S6]
	Iraq:1	[S8]
	Germany: 1	[S46]
	Sweden: 1	[S50]

the titles and abstracts of all retrieved studies. Finally, in the third phase, both reviewers examined the full texts of those studies. Any discrepancies between the two reviewers were handled by discussion with a third reviewer (AA). We used the Rayyan software, which is a web-based systematic review tool, during removal of duplicates, screening, and selection [5].

2.4. Data extraction

Data was extracted and recorded in line with previous similar studies

Table 3
Features of the included corpora.

Characteristics	Studies	Study reference (refer to Table 1)
Corpora features	Manual annotation: 23	[S6], [S7], [S8], [S10], [S12], [S14], [S19], [S20], [S22], [S24], [S26], [S28], [S30], [S31], [S35], [S39], [S45],[S46]–
	Semi-Automatic annotation:7	[S49],[S51]
	Raw text: 18	[S16], [S17], [S25], [S34], [S40], [S43], [S53]
Corpora type	Multipurpose: 21	[S2], [S4], [S9], [S11], [S15], [S18], [S21], [S23], [S27], [S29], [S32], [S33], [S36]–[S38], [S41], [S42],[S50] [S2],[S4], [S8],[S9],[S11],[S19],[S21], [S27],[S33], [S34], [S45], [S15], [S16], [S17], [S29], [S32], [S36], [S37], [S43], [S46],[S51]
	Dialectal: 12	[S6], [S12], [S22], [S30], [S31], [S40], [S42],[S47],[S48],[S52],[S49],[S50]
	Sentiment:10	[S14],[S23],[S25], [S26], [S35], [S39],
	Speech: 3	[S41], [S44],[S53]
	Images: 2	[S10], [S18], [S28] [S7],[S38]
Source of corpus	Twitter:14	[S12], [S22], [S23], [S24], [S25], [S26], [S31], [S35], [S39], [S44] [S48],[S49],[S50],[S53]
	Corpora:10	[S17], [S23], [S29], [S30], [S33], [S37],
	Website:9	[S43], [S45],
	Books:9	[S4], [S8], [S9], [S14], [S16], [S18],
	Facebook:3	[S26], [S31], [S34]
	Newspaper:3	[S4], [S8], [S9], [S11], [S15], [S21],
	Records:2	[S27], [S32], [S33],
	Magazine:2	[S19], [S26], [S31]
	Novels:1	[S2], [S4], [S8] [S46],[S51] [S4], [S11] [S40]

[1], the authors felt this was the best way to organize and display reliable and readable data (Tables). Two reviewers (NM and MS) individually conducted data extraction, and any discrepancies were resolved by a third reviewer (AA). Study characteristics such as author names, year, and country of publication were extracted. Free Arabic corpora-related characteristics were also extracted, including the name, type/purpose, website link, size, and the primary sources of the corpora. It must be noted however that some of the web links were updated in Oct 2021 as the authors found links that were active at the time of study extraction (Jan 2021) had changed. This mainly due to the nature of the links being a short link format. It is perfectly reasonable to expect that links that are currently active at the time of the writing of this review would become inactive in the future.

2.5. Study quality assessment

Scoping reviews differ from systematic reviews; scoping reviews cover a general topic and include more diverse studies. Therefore, scoping reviews do not always assess the quality of related studies [4]. Thus, we did not evaluate the quality metrics of the included studies in this review.

2.6. Data synthesis

We aimed to organize the free Arabic corpora based on their purpose, type, and primary sources, as shown in Tables 1–8, thus the narrative approach was used to synthesize the extracted data. We have composed different taxonomies and tables to summarize the characteristics of the

Table 4
Multipurpose corpora.

Study ref (refer to Table 1)	Name	Size	Available link	Source
S2	N/A	1.5 billion	https://arxiv.org/abs/1611.04033#:~:text=It%20contains%20over%20a%20billion,a%20period%20of%20fourteen%20years.	Five million Arabic newspaper articles
S4	KACST	700 million words	https://www.proquest.com/docview/1697675934?pq-origsite=gscholar&fromopenview=true	Old manuscripts, books, newspapers, magazines, curricula, University theses, websites, refereed periodicals, official prints, and news agencies.
S9	ANT	9, 670, 000 words	https://bit.ly/3uu7q1X	the Tunisian radio website "Jawhara-FM"
S33	ADCC	4 million words	https://bit.ly/2P0FHph	Frequently asked questions on websites, books of learning Arabic, conversations in social media, children's stories, and Alsuliti corpus [S54]
S27	CANER	258,241 words	https://bit.ly/2ZVGzh3	Hadith Sahih Albukhari
S8	PAAD	58,465 words	https://rb.gy/kjvezl	Social network, global forums, ideological websites, and newspaper
S34	ANT corpus v2.1	31,798 document	https://rb.gy/j6dywt	Arab news websites. "AlArabiya"11, "BBC"12, "CNN"13, "France24"14 and "SkyNews"15
S11	RCATS	984 words	https://rb.gy/ddws28	Arabic magazines and books
S45	ANS	N/A	https://github.com/latynt/ans	Arabic news texts (ANT) corpus [S34]
S19	DZDC12	N/A	https://bit.ly/3uqX6bb	Collection of several Facebook comments written in Arabizi–French
S21	Kunuz	N/A	https://bit.ly/3srfgrf	Hadith Sahih Albukhari
S36	Shamela	1 billion words	https://shamela.ws/	Al-Maktaba Al-Shamela
S17	N/A	292 million words	https://bit.ly/3smKyj9	Open Islamicate texts initiative (OpenITI) corpus [S55]
S16	OSIAN	157 million words	https://www.clarin.eu/	International Arabic news websites
S32	Tashkeela	75 million words	https://rb.gy/bzkbrm	Islamic classical books collected from Shamela Library [S56]
S37	N/A	153k sentences 3.5 M tokens	https://paperswithcode.com/dataset/opensubtitles	WIT3: web inventory of transcribed and translated talks [S57]
S29	N/A	100,000 sentences	https://rb.gy/7vgkul	Europarl-v7 English-French parallel corpus [S58]
S43	N/A	32,131 words	https://rb.gy/sf26l8	United Nations corpus [S59]
S15	A7′ta	35,32 words	https://rb.gy/l1blvc	From the Arabic book (Linguistic Error Detector–Saudi Press as a Sample) [S60]
S46	CSCS	11,286 tokens	mohamed.balabel1@ibm.com	Transcript of 5.3 h of recorded speech from Interviews were conducted with 12 participants
S51	WAW	194,927 words	https://alt.qcri.org/resources/	Transcript of recordings from three international conferences, which took place in Qatar

Table 5
Dialectics corpora.

Study ref (refer to Table 1)	Name	Size	Available link	Source
S40	Gumar	110 million words	https://bit.ly/3klJ8Tb	Arabic Form Novels
S6	Habibi	3.5 million words	https://rb.gy/1ojzrc	More than 30,000 Arabic song lyrics in 6 Arabic dialects
S22	Arap-Tweet	2.4 million Tweets	https://bit.ly/3bBsR8P	Twitter
S42	PADIC	231,234 words	https://rb.gy/hj89ix	Movies and TV shows translated to different Arabic dialectal
S31	Curras	55,960 words	https://bit.ly/3byX4VU	Facebook, Twitter, Blogs, and Tv shows
S30	MADAR	47,466 words	https://rb.gy/hvzwlb	Translation of traveling Expression Corpus (BTEC) [S61]
S12	N/A	5361 tweets	https://rb.gy/ilcy5z	Twitter
S47	MADAR CODA	10,000 sentences	https://rb.gy/t8hzg0	MADAR Corpus [S30]
S48	DAICT	5358 tweets	https://rb.gy/nrvsxa	Twitter
S52	N/A	200,000 words	https://rb.gy/gkueoj	Gumar Corpus [S40]
S49	DART	144,596 words	https://rb.gy/ncc1pb	Twitter
S50	Shami	117,805 sentences	https://rb.gy/8yt832	Twitter

included studies. The summary description of each feature is presented in the following sections.

3. Results

3.1. Search results

The initial search of six bibliographic databases returned a total of 1020 citations (Fig. 1). Of which, we found 68 citations that were

Table 6
Sentiment analysis corpora.

Study ref (refer to Table 1)	Name	Size	Available link	Source
S41	N/A	190 million words	https://rb.gy/dlcogs	News articles, 63 thousand consumer reviews, social media, and holy Quran
S23	N/A	150,000 words	https://rb.gy/vea9g7	The Arabic Wikipedia, Arabic Gigaword corpus [S62] and Twitter.
S14	RSAC	81,277 words	https://rb.gy/ost1m0	Hotel reviews from Booking.com website
S25	Moarlex	36,775 words	https://rb.gy/qkpjng	Twitter
S35	AraSenTi	17,573 tweets	https://rb.gy/au2taz	Twitter
S26	MASC	13,760 words	https://rb.gy/uewzqh	Customer reviews from Twitter, Facebook and different Arabic review websites
S39	NileULex	6000 words	https://rb.gy/97ghwa	Twitter
S24	SauDiSenti	4431 words	https://rb.gy/cls3w4	Twitter
S44	ArSenTD-LEV	4000 tweets	https://bit.ly/3bGKnbw	Twitter
S53	N/A	43,304 words	https://rb.gy/wuedne	Twitter

duplicates and were hence removed. Of the remaining 952 articles, 657 articles were unrelated to any language corpora while 50 articles were excluded because they were unrelated to the Arabic corpora. Moreover, 63 articles did not meet the criteria for this scoping review and were therefore excluded. After screening the titles and abstracts, 182 articles remained for full-text reading. Of these, 79 articles were not corpus-related according to the inclusion criteria of this review. Sixty-seven articles were either proposals of Arabic corpus or were in the developmental stage, and these were therefore excluded. Finally, 36 studies were selected after reading the full texts. Four more studies that matched the inclusion criteria were included after backward and forward reference list checking.

According to expert advice, we also searched The International Conference on Language Resources and Evaluation (LREC) as many related papers are available in this database. We retrieved 8 articles from the LREC published between 2015 and 2020. In total, we reviewed 48 articles in this study.

3.2. General description of the studies

All included studies were conducted between 2015 and 2020, as outlined in Table 2. Table 1 contains the study reference codes generated by authors for ease of referring to each study in Tables 2–8. Most studies were reported in 2018 and 2020 (12/48), while a few were published in 2016 (8/48) and 2019 (7/48). Majority of the studies were from Arab countries (23/48), including Saudi Arabia, Qatar and UAE. However, USA and Saudi Arabia jointly rank highest when it comes to individual country of origin of publication (6/48). Of the 48 included articles, 31 were conference papers, indicating that these studies are mostly accepted in conferences. All corpora, except for ten (10/48), had a name.

3.3. Description of the free Arabic corpora

This section describes the characteristics of the included free Arabic corpora. As shown in Table 3, we have classified the type of corpora into five categories based on their primary purpose, namely: multipurpose corpora (21/48), dialectal corpora (12/48), sentiment corpora (10/48), speech corpora (3/48), and images corpora (2/48).

The primary sources of the corpora are divided into nine domains; the most common source was Twitter (10/48), followed by websites (9/48), books (9/48), other corpora (8/48), Facebook (3/48), newspapers (3/48), Records (2/48), magazines (2/48), and novels (1/48). We also considered how the annotation was conducted in each corpus i.e., how the information was added as text for the given purpose. Of the 48 corpora, 15 were annotated manually by expert annotators, seven were annotated semi-automatically using various tools, and 17 were reported as raw text as they did not provide any type of annotation and were only available as text files. Each type of corpus is described in the following sections.

3.3.1. Multipurpose corpora

Multipurpose corpora are those created to serve large-scale applications such as NLP and IR (Information Retrieval) applications. This includes machine translation, sentiment analysis, emotion detection, abusive message detection, gender identification, dialect identification, and segmentation. Table 4 illustrates the details of each free Arabic corpora including its name, size, source, and the accessible link.

3.3.2. Dialectal corpora

Dialectal corpora comprise the most factual data and knowledge of

Table 8
Image corpora.

Study ref (refer to Table 1)	Name	Size	Type	Available link	Source
S38	SmartATID	16,472 images	Arabic printed and handwriting text image	https://cutt.ly/SlEZqiS	Arabic/ Latin Text Images Database for recognition research (ALTID) [S64]
S7	ASAYAR	1765 images	Arabic multilingual scene text	https://bit.ly/3bCd0qy	From videos of two round trips in Moroccan highways

Table 7
Speech corpora.

Study ref (refer to Table 1)	Name	Size	Type	Available link	Source
S10	ADI17	3033 h	Dialect Identification	https://bit.ly/3kon1vo	Consists of videos from 17 Arabic countries
S28	ARAB-ANDALUSIAN	125 h	Moroccan tradition of Arab-Andalusian music	https://bit.ly/3usZQVk	Collection of the Arab-Andalusian corpus consists of more than 125 h of music
S18	Arpod	8.1 h	Speech-based multi-lingual and multi-dialectal identification tasks	https://rb.gy/ttyyub	Speech data from “Arab podcast” website3

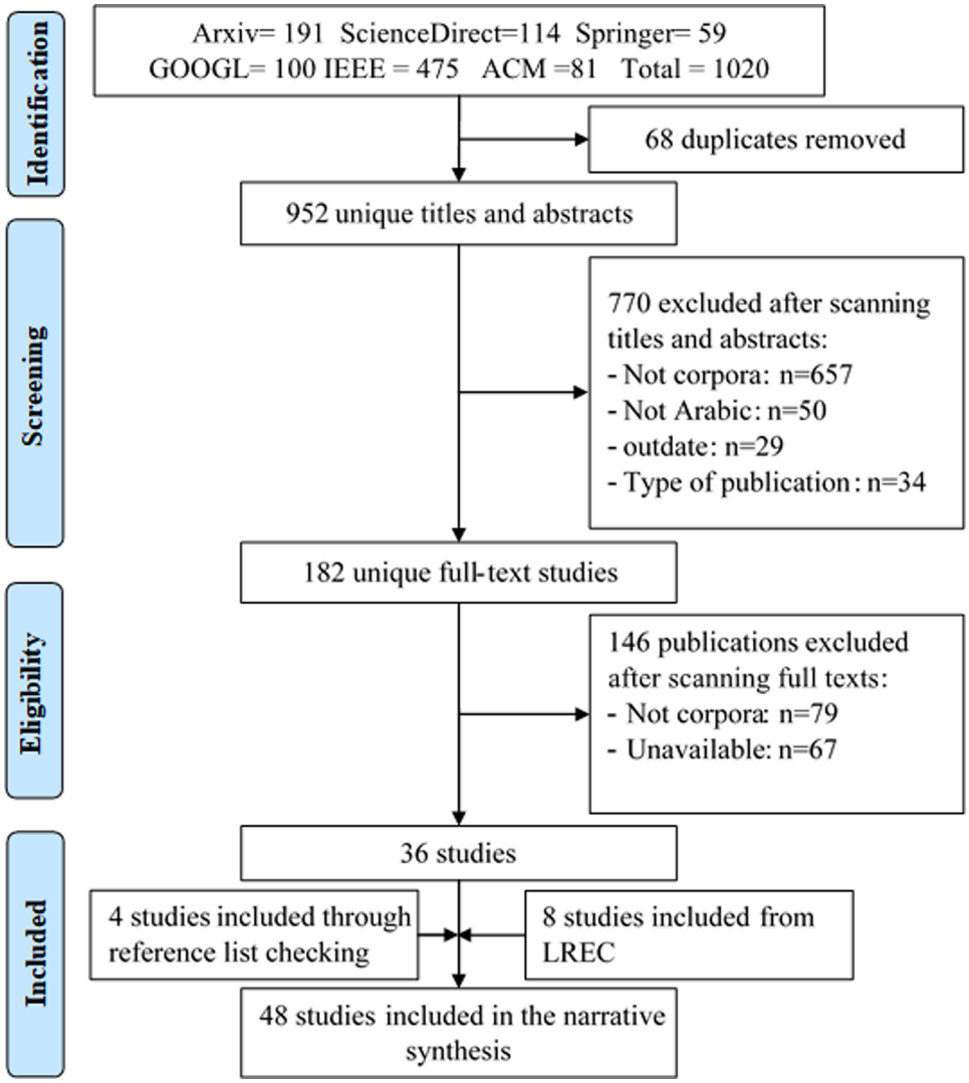

Fig. 1. PRISMA flow chart.

dialect-based differences that are investigated according to a particular region, with experimental details and valid authenticity. Table 5 provides elements that belong to the Arabic dialectal corpora, developed to serve Arabic dialogue and the modern Arabic language.

3.3.3. *Sentiment analysis corpora*

Sentiment Analysis Corpora (Opinion Mining) consist of data concerning the feelings expressed by internet users regarding a particular topic. The annotations that include these feelings are collected and parsed, and several tasks are performed on them. Table 6 demonstrates the features of sentiment free Arabic corpora including their name, size, source, and link.

3.3.4. *Speech corpora*

Speech corpus is an extensive collection of audio records in the Arabic language. Speech corpora also possess additional documents or text files, including transcriptions of spoken words and the number of times each word is transpired in the records. In this review, we found that only three Arabic speech corpora were available for public use, as shown in Table 7.

3.3.5. *Image corpora*

Image corpora are printed or handwritten Arabic documents or databases containing images, and can mainly be used for document analysis scenarios, Arabic text recognition, and evaluating the quality of the image that contains Arabic writing. Table 8 shows two image corpora that were included in this review.

4. Discussion

To the best of our knowledge, this is the first scoping review that followed the PRISMA-ScR guidelines to identify freely available Arabic corpora, thereby providing insightful information for readers and researchers in this field. Although a previous study provided invaluable and unprecedented information for those seeking free Arabic corpora [1], it was largely based on a survey conducted on 20 participants who identified 26 sources. The author further identified 40 resources and extracted author information and other metrics where available. A total of 66 resources were then categorised by the type of corpora and explained further by highlighting important metrices like the number of words (size of corpora) and their origins. Most of the corpora (21/48) included in this review were for multiple purposes such as NLP and IR applications.

The most popular databases were searched to retrieve literature in this field. We also minimised publication bias by conducting backward and forward reference checking from the retrieved studies using Google Scholar. In addition, two independent reviewers were assigned for study selection and data extraction to further avoid selection bias. As this

scoping review identified freely available Arabic corpora from the existing literature, its possible unpublished corpora were missed. Furthermore, the search was restricted to studies published within the English language, since this review was about freely available Arabic corpora, we likely have missed quality papers published in the Arabic language. Finally, the included papers were published between 2015 and 2020; therefore, studies published before that period may have also been excluded.

Corpora play a very important role in training machine learning systems to build various NLP tools. We believe that having access to this scoping review and other similar studies will promote the access to freely available resources, especially for a language such as Arabic with its multiple dialects. Moreover, such extensive reviews will avoid the duplication of efforts especially with the high costs of building corpora and the limited resources available for many research institutes in the Arab world and around the globe. Currently, the various resources available for Arabic corpora are scattered in multiple locations and some are not maintained anymore. We believe that this review will enhance the research outcomes in Arabic NLP and its multiple dialects, and we hope that the research community benefits from this review and advances in Arabic NLP research. We also aim to encourage researchers to develop new Arabic datasets concerning areas that are yet to be covered to complement currently available resources. In addition, as this scoping review focused on freely available resources, we believe that making more datasets freely available would be extremely beneficial as paid corpora could generate a limited impact given the lack for funding, especially for young scholars. Finally, readers must bear in mind and expect that links to corpora reported that are currently active at the time of the writing of this review could well become inactive in the future, this merely further highlights challenges faced by researchers when it comes to accessibility.

5. Conclusion

This scoping review identified freely available corpora in the Arabic language. We used PRISMA guidelines to ensure that only peer-reviewed database searches were included. Additionally, we included corpora from LREC, a popular language source, after seeking advice from experts in the field. Our review provides valuable insights in this field whilst highlighting the scarcity of corpora in Arabic compared to other language such as English. Especially considering its global usage by more than 1.5 billion Muslims and it being the native language of over 250 million people as the Quran, the core text of Islam, is written in Arabic. Furthermore, this study is the first to compile various types of freely available Arabic corpora as previous corpora-based studies have shown scattered availability on various websites that are often not maintained. Through this review, we aim to initiate advancements in Arabic NLP research, to encourage researchers in building new Arabic datasets on areas that are currently uncovered, and encourage corpora to be made freely available and more accessible to young researchers, enthusiasts, and scholars. Ultimately encouraging more AI driven projects using NLP for the Arabic language.

Declarations

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Availability of data and material

N/A

Code availability

N/A

Author's contributions

Nashva Ali & Arfan Ahmed developed the protocol and conducted the search with guidance from and under the supervision of Alaa Abdalrazaq & Mowafa Househ. Study selection and data extraction were carried out independently by Mahmood Alzubaidi. Arfan Ahmed drafted the manuscript, and it was revised critically for important intellectual content by all authors. All authors approved the manuscript for publication and agree to be accountable for all aspects of the work.

Declaration of Competing Interest

The authors have no competing interests to declare.

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